

# MATH1231 CALCULUS

## Chapter 4D

Term II 2026

A/Prof Mark Holzer

UNSW Sydney  
mholzer@unsw.edu.au

June 1, 2026

## (4.6) Taylor Series

### Definition

- Suppose that  $f(x)$  is infinitely differentiable at  $x = a$ . Then

$$P(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x - a)^k$$

is the **Taylor Series** of  $f(x)$  about  $a$ .

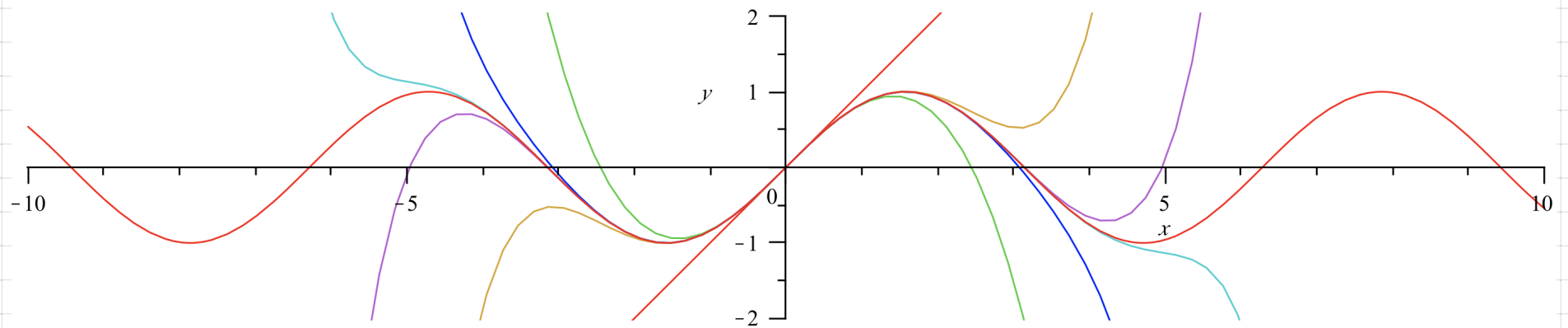
- The **Taylor Series** of  $f(x)$  about  $x = 0$  is also called a **Maclaurin series**.

Note: The  $n$ th partial sum of the Taylor Series is the  $n$ th Taylor polynomial  $P_n(x)$ .

## Example

Find the Maclaurin series for  $\sin x$

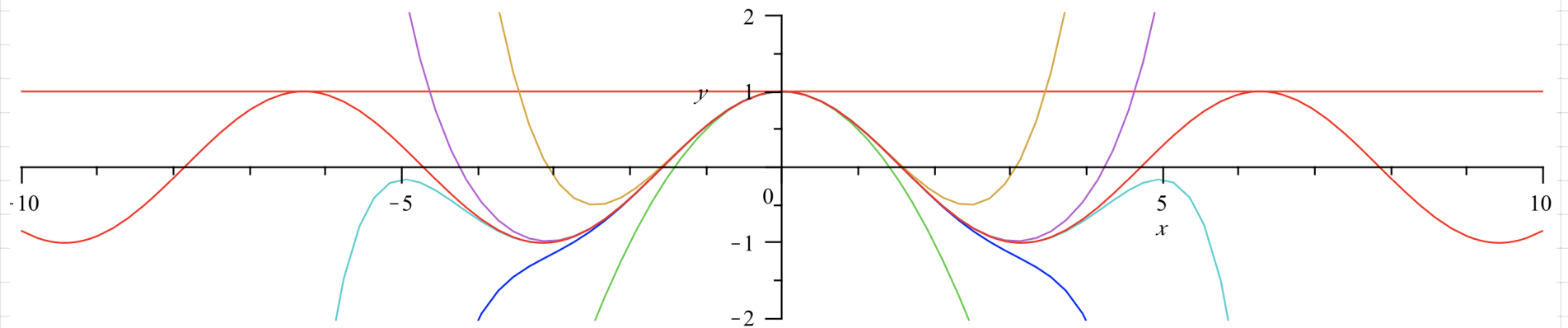
$\sin x$  and the first few partial sums of its Maclaurin series



## Example

Find the Maclaurin series for  $\cos x$

$\cos x$  and the first few partial sums of its Maclaurin series



## Example

Find the Maclaurin series for  $\log(1 + x)$

## Example

Find the Taylor series for  $\log(x)$  about  $x = 1$ .



# Convergence

The  $n$ th partial sum of the Taylor series for  $f(x)$  is the  $n$ th-order Taylor polynomial  $P_n(x)$ . Taylor's theorem provides an expression for the remainder  $R_{n+1}(x) = f(x) - P_n(x)$ .

The sequence of partial sums, and hence the Taylor series, converges to  $f(x)$  if  $\lim_{n \rightarrow \infty} R_{n+1}(x) = 0$ .

We can prove convergence if we can bound the derivative term in the Lagrange remainder formula for  $R_{n+1}$ :

$$|R_{n+1}(x)| = \left| f^{(n+1)}(c) \frac{(x-a)^{n+1}}{(n+1)!} \right| \leq M \frac{|x-a|^{n+1}}{(n+1)!}$$

where  $M$  is an upper bound such that  $|f^{(n+1)}(t)| \leq M$  for  $t$  between  $x$  and  $a$ . If such an upper bound  $M$  can be established for all  $n$  (larger than some finite  $N$ ), then  $\lim_{n \rightarrow \infty} R_{n+1}(x) = 0$  and the Taylor series converges to  $f(x)$ .

# Common Maclaurin Series

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k \quad x \in (-1, 1)$$

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \quad x \in \mathbb{R}$$

$$\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} \quad x \in \mathbb{R}$$

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} \quad x \in \mathbb{R}$$

$$\cosh x = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!} \quad x \in \mathbb{R}$$

$$\tan^{-1} x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{2k+1} \quad x \in [-1, 1]$$

Note the convergence restrictions on  $x$ .

## Example

Find the Taylor series for the function  $\cos x$  about the point  $x = \pi/2$ .  
On what interval does the Taylor series converge to  $\cos x$ ?

## Example

Prove that the Taylor series for  $e^x$  about  $x = 0$  converges to  $e^x$  for all  $x$ .

# Example/Application

Evaluate  $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ .

## Example/Application

Express  $\int_0^1 \sin(x^2) dx$  as a infinite series (there is no explicit antiderivative).

## (4.7) Power Series

### Definition

- ▶ A series of the form

$$\sum_{k=0}^{\infty} c_k (x - a)^k ,$$

where  $\{c_k\}$  are constant real coefficients, is called a **power series**.

- ▶  $a$  is called the **centre** of the power series.
- ▶ A Taylor series is a special case of a power series.

We will see that *convergent* power series obey the rules of algebra and can be added, subtracted, and multiplied. They are also easily differentiated and integrated term-by-term. We therefore need to study the convergence properties of power series.

Consider the power series

$$F(x) = \sum_{k=0}^{\infty} \frac{1}{2^k} x^k .$$

Note that  $F(0) = 1 + 0 + 0 + \dots = 1$  and  $F(1) = 1 + 1/2 + 1/4 + \dots = 2$ . However,  $F(3) = \sum_{k=0}^{\infty} (3/2)^k$  is a divergent geometric series.

How do we determine the values of  $x$  for which a given power series converges?



# Radius of Convergence

The convergence of a power series is characterized by its **radius of convergence**.

The radius of convergence is a positive number  $R$  such that the series diverges for  $x$  with  $|x - a| > R$ , but converges absolutely for  $x$  with  $|x - a| < R$ . The series may or may not converge at either of the end points  $x = a - R$  and  $x = a + R$ .

$R = \infty$  if the series converges for all  $x$ .

$R = 0$  if it converges only for  $x = a$ .

## Finding $R$

We can find  $R$  by testing for absolute convergence with the ratio test:

Note: If  $R$  is finite, need separate test for the end points. Use comparison, integral, alternating/Leibnitz test as needed.

## Example

Establish the convergence properties of  $F(x) = \sum_{k=0}^{\infty} \frac{1}{2^k} x^k$

## Example

Discuss the convergence of the power series  $F(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

## Example

Discuss the convergence of the power series  $F(x) = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} (x - 5)^n$

## Example

Discuss the convergence of the power series  $F(x) = \sum_{n=1}^{\infty} n x^n$

## Example

Find the largest open interval on which  $F(x) = \sum_{k=2}^{\infty} \frac{(3x-2)^k}{k \ln k}$  converges.

## (4.8) Manipulation of Power Series

Functions defined by convergent power series can be added, subtracted, multiplied, differentiated, integrated ...

### Differentiation and Integration of Power Series

Suppose that  $f(x) = \sum_{k=0}^{\infty} a_k(x-a)^k$  is a convergent powers series for  $|x-a| < R$ .

- $f(x)$  is **continuous** and **differentiable** for  $|x-a| < R$

$$\frac{df}{dx} = \sum_{k=0}^{\infty} \frac{d}{dx} \left( a_k(x-a)^k \right) = \sum_{k=1}^{\infty} k a_k (x-a)^{k-1}$$

- $f(x)$  is **integrable** on  $|x-a| < R$

$$\int f(x) dx = \sum_{k=0}^{\infty} \int \left( a_k(x-a)^k \right) dx = \sum_{k=0}^{\infty} \frac{a_k}{k+1} (x-a)^{k+1} + C$$

**convergent power series** can be **differentiated** and **integrated term by term**.



## Example

Establish a power series for  $\log(1+x)$  with  $x \in (-1, 1)$  by using the

power series  $\sum_{n=0}^{\infty} t^n, \quad -1 < t < 1.$

## Example

$$\begin{aligned}f(x) &= \sum_{k=0}^{\infty} \frac{x^k}{k!} \\ \Rightarrow f'(x) &= \sum_{k=0}^{\infty} \frac{kx^{k-1}}{k!} = \sum_{k=1}^{\infty} \frac{kx^{k-1}}{k!} = \sum_{k=1}^{\infty} \frac{x^{k-1}}{(k-1)!} \\ &= \underbrace{\sum_{n=0}^{\infty} \frac{x^n}{n!}}_{n=k-1} = \underbrace{\sum_{k=0}^{\infty} \frac{x^k}{k!}}_{k=n} = f(x)\end{aligned}$$

Note too that  $f(0) = 1$ .

But  $f'(x) = f(x)$  with  $f(0) = 1$  has the unique solution  $f(x) = e^x$

$$\Rightarrow e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$